Satellite Video Object Detection Based on Enhanced 3DTV Regularization and Gaussian Prior

Qian Yin, Wei An, Ting Liu, Yang Sun, Zaiping Lin, Yulan Guo, *Senior Member, IEEE*, and Hanyun Wang

***Abstract*— Satellite videos have played important roles in many applications in recent years due to the advantages of continuous providing high temporal resolution remote sensing images. Although much progress has been achieved for moving object detection (MOD) in satellite videos, the low-rank characteristics of background and the intensity variations of moving objects across frames have not been fully exploited. In this article, we propose an efficient method for MOD in satellite videos, which models the background with enhanced 3-D total variation (E-3DTV) regularization and the moving objects with Gaussian prior. Specifically, considering that the gradient maps on the spatial and temporal dimensions exhibit different physical meanings, we model the background with different Laplacian sparsity priors for the gradient maps along the spatial and temporal dimensions for 3DTV regularization. Different from current methods, which model moving objects with sparsity characteristics in each frame alone, we utilize Gaussian prior to model intensity changing characteristics of moving objects across frames. After integrating background model and moving object model into low-rank sparse matrix factorization framework, the alternating direction method of multipliers (ADMM) is adopted to iteratively optimize the parameters of background and moving object models. We conduct experiments on VISO and SkySat datasets, and the results demonstrate that our method achieves superior MOD performance with high computational efficiency compared to state-of-the-art methods.**

***Index Terms*— 3-D total variation, background modeling, Gaussian prior, low-rank matrix factorization (LRMF), moving object detection (MOD), satellite videos.**

I. Introduction

SATELLITE videos are collected by video satellite constellations and are widely applied in many remote sensing applications, such as environment monitoring [1], [2], urban monitoring [3], [4], and traffic condition monitoring [5], [6], [7], [8]. Due to the characteristics and limitations of optical satellite imaging sensors, the MOD in satellite videos faces significant challenges, including small sizes of objects, uneven moving of background, and low contrast to the complex and dynamic background. These characteristics make the slowly moving objects difficult to distinguish from background clutters and also lead to inconsistent detection, such as missing or partial detections. Thus, the MOD in satellite videos with strong robustness is still a challenging task.

Recently, many techniques have been explored for MOD in satellite videos [9], [10], [11], [12], [13], [14] and can be roughly divided into frame differencing-based methods, background modeling-based methods, and learning-based methods. The frame differencing-based methods utilize the image appearance features and the estimated motion cues between adjacent images to conduct object detection in satellite videos [15], [16]. Learning-based methods utilize well-designed network architectures to extract appearance feature and spatiotemporal feature for moving object detection (MOD) in satellite videos [17], [18], [19], [20], [21].

The background modeling-based methods establish a representative background model by statistically analyzing the background characteristics within videos and then extract the foreground objects by subtracting the estimated background model from the current frame. In particular, the background in satellite videos is usually stable and similar across frames and exhibits low-rank characteristics. In contrast, moving objects usually occupy only a small proportion of pixels compared to the background and exhibit sparse characteristics. Thus, low-rank and sparse decomposition (LRSD) methods treat background as a low-rank matrix and moving objects as a sparse matrix and take satellite videos as the summation of background and moving objects. This problem can be solved by the robust principal component analysis (RPCA) algorithm [22], [23], [24], [25], [26], [27], and many solutions have been proposed in recent years [9], [10], [28], [29].

These LRSD-based methods typically use convex or nonconvex rank functions to explore and encode spatial prior structures underlying satellite videos. Then, the background of the image can be accurately estimated. Based on the closed-form solution, the detection results can be obtained via iterative optimization. Furthermore, to enhance background suppression capabilities and improve the detection accuracy, some LRSD-based methods incorporate 3-D total

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variation (3DTV) regularization or its variants [30], [31] to enforce local smoothness prior on background. Usually, existing 3DTV methods assume that the sparsity of gradient maps calculated on the spatial and temporal dimensions conform to the same identical and independent distributed Laplacian prior distribution. However, gradient maps on the spatial and temporal dimensions usually have different physical meanings. Meanwhile, it requires a large amount of computations for singular value decomposition (SVD) and complex regularization, which is difficult to meet the practical efficiency requirement of MOD. Moreover, LRSD-based methods assume that objects are sparse within a single frame and usually use ℓ_1 norm to constraint foreground. However, this assumption ignores the intensity variation characteristics of moving objects across frames.

To solve these problems, we propose an efficient method for MOD in satellite videos based on enhanced 3-D total variation (E-3DTV) regularization and Gaussian prior. Specifically, to model background in satellite videos, gradient maps from the spatial and temporal dimensions are modeled with different Laplacian sparsity priors for 3DTV regularization. Spatial gradient maps reflect the changes of local grayscale intensity in the spatial domain, with their amplitudes indicating the strength of high-frequency information such as edges and textures. Temporal gradient maps describe intensity variations between adjacent frames in a video sequence. By constraining the sparsity of gradient maps, the E-3DTV can reduce the redundant gradient information to avoid oversmoothing and enhance the edge to preserve the structure details, which is critical for accurate background reconstruction in satellite videos. Moving objects exhibit temporally coherent intensity changes (consistent direction/speed). By statistically analyzing intensity variations across frames, we observe that these variations approximately follow a Gaussian distribution. While noise pixels occasionally exhibit large (but meaningless) intensity jumps, the Gaussian prior effectively penalizes these uncorrelated fluctuations. Thus, for moving objects in satellite videos, we use the Gaussian distribution to model the intensity changes of moving objects and employ ℓ_2 regularization to better account for the characteristics of moving objects in our optimization process. Through this manner, we can more effectively capture the temporal dynamics of moving objects, which would otherwise be neglected in traditional sparse decomposition methods. By combining E-3DTV-based background modeling and Gaussian prior-based foreground modeling, we take the MOD task in satellite videos as an LRSD problem in this article and adopt the alternating direction method of multipliers (ADMMs) to iteratively optimize the parameters of background and moving object models. Experimental results conducted on the VISO and SkySat datasets demonstrate that our method achieves superior MOD performance with high computational efficiency compared to state-of-the-art methods.

The main contributions of this article are listed as follows.

1) We model the background in 3DTV with different Laplacian sparsity priors for the spatial and temporal gradient maps and model the intensity changing characteristics of moving objects across frames with Gaussian prior.
2) We introduce a unified framework to integrate the E-3DTV regularization and Gaussian prior and propose a closed-form solution to optimize the reconstruction model and improve the robustness of object detection in dynamic scenes.
3) With the help of better exploration of background and motion prior, our method has advantages compared with state-of-the-art methods in detecting moving objects quickly and suppressing background clutter effectively.

II. Related Work

In this section, we briefly review the state-of-the-art works for multiple MOD in satellite videos, which can be roughly divided into frame differencing-based methods, background modeling-based methods, and learning-based methods.

A. Frame Differencing-Based Methods

The main idea of frame differencing-based methods is to explore the motion cues between consecutive frames for MOD in satellite videos. To exploit the motion cues, the changes between consecutive frames are used to obtain the region proposals of interesting objects [15], [16]. Specifically, Ao et al. [15] detected moving objects by computing the differences between adjacent frames and then performed segmentation to obtain the detection results. To solve the problem of high false alarms caused by the complex background changes in the frame difference-based methods, Yin et al. [16] proposed a motion modeling baseline to improve the recall performance and reduce the false alarms based on accumulative multiframe differences and robust matrix completion. However, the detection performance would be degraded by the sudden change in dynamic backgrounds.

B. Background Modeling-Based Methods

The main idea of background modeling-based methods is to construct a background model that can effectively describe the characteristics of nonmoving (i.e., static) backgrounds. Current background modeling methods for satellite videos can be categorized into two groups, i.e., statistical modeling methods [5], [32] and subspace learning methods [9], [10], [28], [29]. The statistical modeling methods first generate a background reference by statistically analyzing the intensity distribution of image pixels and then detect moving objects by comparing the input image with background reference. Kopsiaftis and Karantzalos [5] compared the changes between each pixel and its surrounding pixels within two frames and then determined whether the pixel belongs to the background. Subsequently, Ahmadi et al. [32] utilized the median background model to create a reference (background) frame and obtained detection results by subtracting each frame with the reference background frame. However, these methods are sensitive to illumination changes and background variations.

The subspace learning methods assume that background frames are temporally stable and similar and thus can be assumed to lay in a low dimensional subspace, which is commonly estimated using low-rank matrix decomposition. Zhang et al. [9] first converted the MOD task in satellite videos

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into a low-rank matrix recovery problem and modeled the foreground with a structured sparse matrix and the background with a low-rank matrix. To remedy the delay problem in the method [9], Zhang et al. [28] proposed an online low-rank and structured sparse decomposition method by reformulating the batch-based low-rank and structured sparse decomposition as the matrix factorization approximation of nuclear norm. Later, Zhang et al. [10] proposed a moving-confidence-assisted matrix decomposition model to suppress false alarms caused by the local misalignment and local intensity variation. Subsequently, Yin et al. [29] incorporated the idea of LRSD into a tensor framework with bounded noise and proposed a generalized weighted Schatten p-norm minimization to separate moving objects from background. Although these LRSD methods improve detection efficiency, most of them usually combine with complex regularization terms to improve detection capability, which decreases computational efficiency.

C. Learning-Based Methods

With the powerful feature representation capacities, learning-based methods have been successfully applied for object detection in satellite videos [17], [18], [19], [20], [21] and also achieved promising performance. LaLonde et al. [17] proposed a two-stage spatial–temporal convolutional neural network (CNN) to effectively and efficiently extract both the motion and appearance features in satellite videos. Zhang et al. [19] presented a weakly supervised method for MOD in satellite videos, where supervised information is replaced by the motion knowledge across frames. Xiao et al. [18] proposed a 2-D backbone to extract static context information from a single frame and a lightweight 3-D backbone to extract dynamic motion cues from consecutive frames. Pi et al. [20] utilized the weighted element-wise multiplication between pairwise differencing images to suppress dynamic clutters. Li et al. [14] and Xu et al. [12] built the largest scale satellite video dataset supporting multiple tasks and object categories, which significantly advances research in this field. Meanwhile, Li et al. [13] presented a systematic review and quantitative analysis of the results published over the last decade, intending to further promote the development of various intelligent processing tasks for satellite videos. Zhao et al. [11] proposed a satellite video moving vehicle detection network, consisting of the historical frame differential module (HFDM) and the improved YOLOv5 algorithm. Additionally, Xiao [21] introduced a deep background prior to model-based methods for moving vehicle detection in satellite videos.

Although learning-based methods have achieved promising performance, their performance relies heavily on large-scale labeled data. However, due to the extremely small size of moving objects and complex backgrounds, annotating moving objects in satellite videos is labor-intensive and time-consuming.

III. Preliminaries

In this section, we first review the low-rank and sparse component decomposition model and 3-D total variation for MOD in detail and then introduce the foundation of low-rank matrix factorization (LRMF).

A. MOD Based on LRSD and 3-D Total Variation

Because of the low-rank characteristics of background and sparse characteristics of moving objects in satellite videos, LRSD-based methods model the satellite videos as the summation of background and moving objects. Given a satellite video $\mathcal{O} \in \mathbb{R}^{H\times W\times L}$, where H, W, and L denote the size of spatial height, width, and the number of frames in the video, respectively, according to the basic idea of LRSD-based methods, the problem of MOD in satellite videos can be formulated as

$$\mathcal{O} = \mathcal{B} + \mathcal{T} + \mathcal{N} \tag{1}$$

where $\mathcal{O}$, $\mathcal{B}$, $\mathcal{T}$, and $\mathcal{N} \in \mathbb{R}^{H\times W\times L}$ denote the original image tensor, background image tensor, moving object image tensor, and noise image tensor, respectively. For MOD in satellite videos, the core of background modeling-based methods is to accurately estimate background image tensor $\mathcal{B}$, and moving object image tensor $\mathcal{T}$ is then obtained by subtracting the estimated background from the original image tensor.

Due to the slow change over time and significant overlap among different frames in satellite video images, background tensor $\mathcal{B}$ conforms to low-rank property [9], [28], [29], which is formulated as

$$\text{rank}(\mathcal{B}) \le r \tag{2}$$

where r is a constant to describe the complexity of the background image and rank$(\cdot)$ represents the rank of a tensor. In addition, moving objects usually occupy only a few pixels in the whole satellite video image, and thus, moving object tensor $\mathcal{T}$ conforms to sparsity prior, which can be described as

$$\| \mathcal{T} \|_0 \le \sigma \tag{3}$$

where σ is an integer determined by the number and size of the small objects. The noise $\mathcal{N}$ is usually modeled with an additive white Gaussian function. By denoting noise intensity as γ, the Frobenius norm of noise image tensor satisfies the condition $\|\mathcal{N}\|_F \le \gamma$.

Generally, an LRSD-based framework for MOD in satellite videos can be solved by replacing $\| \mathcal{T} \|_0$ with $\| \mathcal{T} \|_1$ [33]. Therefore, the LRSD model can be formulated as

$$\begin{aligned} &\min_{\mathcal{B},\mathcal{T}} \|\mathcal{B}\|_* + \lambda\|\mathcal{T}\|_1 \\ &\text{s.t. } \|\mathcal{O} - \mathcal{B} - \mathcal{T}\|_F \le \gamma \end{aligned} \tag{4}$$

where λ denotes the positive regularization parameter and $\|\cdot\|_*$ represents nuclear norm, which is a nonconvex approximation of rank$(\mathcal{B})$. Because of the same weight for all singular values, the nuclear norm minimization method cannot preserve the edge structures of background and thus cannot accurately estimate background.

Due to the superiority of 3DTV on characterizing local smoothness prior of images and preserving edge information effectively, some LRSD-based methods [30], [31], [34], [35] introduce TV regularization or its variants to obtain better background reconstruction ability. Specifically, for a given video sequence $\mathcal{O} \in \mathbb{R}^{H\times W\times L}$, we denote the difference operation calculated along the ith dimension on a video

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sequence $\mathcal{O}$ as D_i, and the gradient map tensor $\mathcal{G}_i \in \mathbb{R}^{H\times W\times L}$ is obtained according to the following equation:

$$\mathcal{G}_i = D_i(\mathcal{O}), \quad i = 1, 2, 3 \tag{5}$$

where $i = 1, 2$ corresponds to spatial height and width dimension respectively, while $i = 3$ corresponds to temporal dimension.

Then, the definition of 3DTV regularization is as follows:

$$\|\mathbf{O}\|_{3\text{DTV}} = \sum_{i=1}^{3} \|\mathbf{G}_i\|_1 \tag{6}$$

where $\mathbf{G}_i$ denotes the unfolded gradient map tensor $\mathcal{G}_i$.

Inherently, 3DTV regularization assumes that the sparsity of gradient maps calculated on the temporal, spatial width, and height dimensions conforms to the same identical and independent Laplacian prior distribution. However, this kind of assumption is far from the real-world cases. We can observer that the sparsity of gradient maps of the spatial dimensions is usually correlated but different from the counterpart of temporal dimension in real-world scenes. Additionally, 3DTV regularization is imposed on the whole image, which results in high computational complexity and slow detection speed.

B. Low-Rank Matrix Factorization

LRMF is designed to approximate a low-rank matrix $\mathbf{P}$ by the product of two matrices with lower rank. This method efficiently captures the essential structure of data with dimensionality reduction and thus can be used to reduce the computational complexity of 3DTV. The LRMF model is formulated as

$$\mathbf{P} = \mathbf{U}\mathbf{V}^{\mathbf{T}} \tag{7}$$

where $\mathbf{P} \in \mathbb{R}^{m\times n}$ represents input data matrix to be approximated; $\mathbf{V} \in \mathbb{R}^{n\times r}$ is the endmember matrix or dictionary, which represents the basis vectors in the reduced-dimensional space; $\mathbf{U} \in \mathbb{R}^{m\times r}$ denotes the abundance or coefficient matrix, which represents the coefficients of matrix $\mathbf{P}$ with respect to the basis matrix $\mathbf{V}$; and r is the rank of the approximation, which is typically much smaller than m and n. In (7), the coefficients in the matrix $\mathbf{U}$ represent how each row of $\mathbf{P}$ can be expressed as a combination of the basis vectors in $\mathbf{V}$. Hence, $\mathbf{U}$ is often referred to representative coefficient matrix. Meanwhile, r constrains the rank of matrix $\mathbf{P}$ and controls the complexity of the factorization. By adjusting r, the low-rank characteristics of $\mathbf{P}$ are exploited, and computational complexity is reduced compared with methods like SVD. By leveraging low-rank approximation, LRMF effectively captures the underlying structure of the data and makes it computationally feasible for large-scale datasets.

IV. Method

A. Overview

In this article, to fully exploit the effectiveness of different low-rank characteristics of background along the spatial and temporal dimensions and the intensity changing characteristics of moving objects across frames in satellite videos, we propose to utilize the E-3DTV regularization and Gaussian prior to model the background and moving objects, respectively. The overall architecture of our method is shown in Fig. 1. Specifically, we treat the MOD task in satellite videos as an LRSD problem and integrate the background model and the moving object model into the LRMF framework. Then, we adopt the ADMM to iteratively optimize the parameters of the background and moving object models.

B. Background and Moving Object Modeling Based on E-3DTV and Gaussian Prior

1) Background Modeling: To fully exploit the advantages of effectively capturing the underlying structure of the data in LRMF and preserving the structural details in 3DTV, we propose to reconstruct the background of satellite videos by conducting LRMF on the gradient maps. Moreover, considering that the gradient maps along the spatial dimension and temporal dimension have different physical meanings, we further model these two kinds of gradient maps with different Laplacian sparsity priors by setting different numbers of ranks in 3DTV for the spatial and temporal gradient maps.

Inspired by the literature [9], [10], [28], [29], the satellite video $\mathbf{O}$ and the gradient map $\mathbf{G}_i$ can be rationally assumed to be low rank. As mentioned in Section III-B, $\mathbf{G}_i$ can be expressed as follows:

$$\begin{aligned} \mathbf{G}_1 &= \mathbf{U}_1\mathbf{V}_1^{\mathrm{T}} \in \mathbb{R}^{HW\times r_1} \\ \mathbf{G}_2 &= \mathbf{U}_2\mathbf{V}_2^{\mathrm{T}} \in \mathbb{R}^{HW\times r_1} \\ \mathbf{G}_3 &= \mathbf{U}_3\mathbf{V}_3^{\mathrm{T}} \in \mathbb{R}^{HW\times r_2} \end{aligned} \tag{8}$$

where r_1 and r_2 represent the rank of satellite videos along the spatial and temporal dimensions, respectively.

Similar to [36] and [37], due to the strong local smoothness prior and sparsity of coefficient matrix $\mathbf{U}$, in this article, we impose 3DTV regularization on the coefficient matrix $\mathbf{U}$ instead of gradient map $\mathbf{G}$ to encode the local smoothness and reduce computational complexity.

Thus, the definition of E-3DTV regularization can be formulated as

$$\begin{aligned} &\|\mathbf{B}\|_{\text{E}-3\text{DTV}} = \sum_{i=1}^{3} \|\mathbf{U}_i\|_1 \\ &\text{s.t. } \mathbf{G}_i = \mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}, \quad \mathbf{V}_i^{\mathrm{T}}\mathbf{V}_i = \mathbf{I}. \end{aligned} \tag{9}$$

2) Moving Object Modeling: Previous methods usually utilize the Laplacian distribution prior (ℓ_1 regularization) to model the moving object, which ignores the intensity variation properties of moving objects across frames. Intuitively, the intensity values of the same pixels of the background across consecutive frames should be the same, while the intensity values of the pixels of noises and moving objects should change. Furthermore, the intensity variations of noisy pixels are irregular across frames, while the intensity variations of moving object are regular. Fig. 2 illustrates the histogram of intensity values of the centroid of a moving object across all frames in satellite videos, and we can observe that the amplitude of the histogram exhibits a Gaussian distribution.

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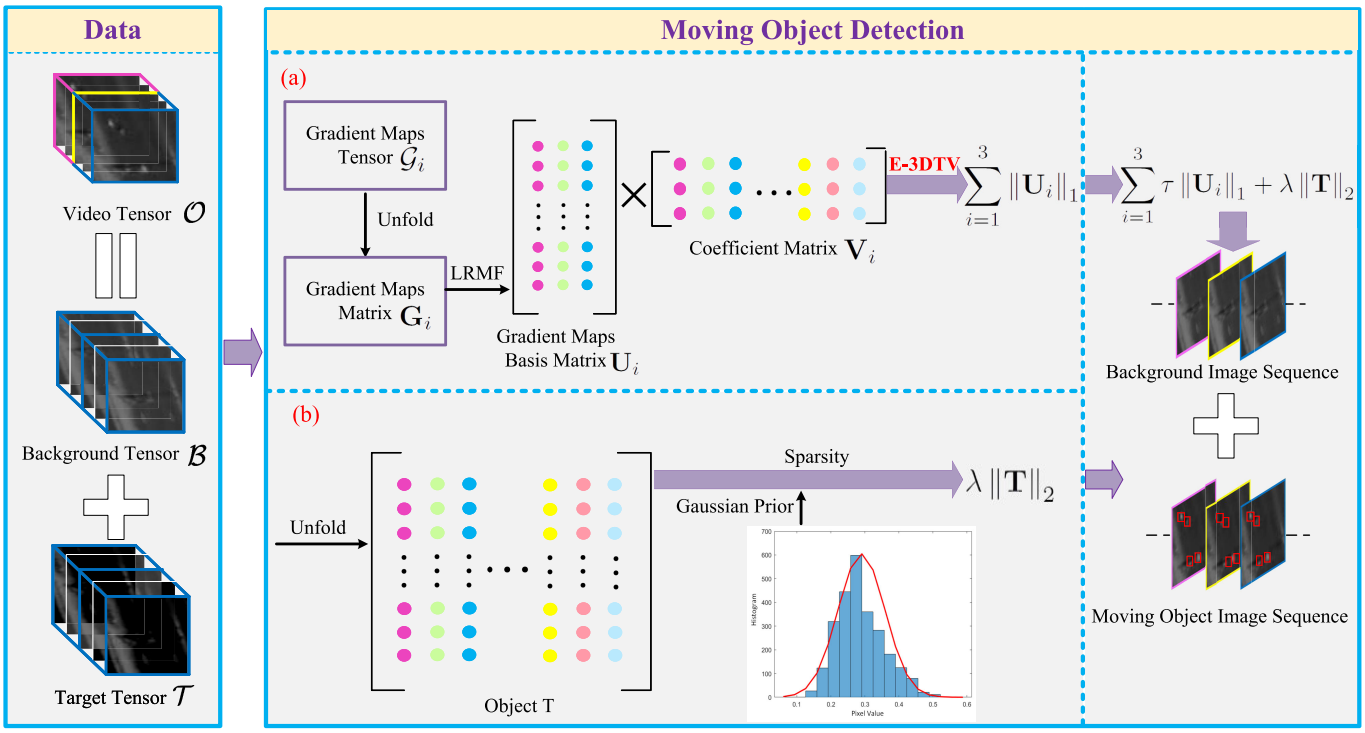


Fig. 1. Framework of our proposed method.

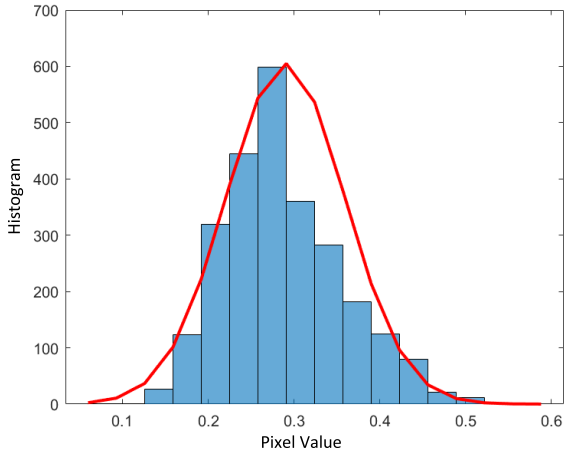


Fig. 2. Illustration of intensity changing characteristics of moving objects.

Inspired from this observation, in this article, we assume that the change probability distribution of intensity values for moving objects in satellite videos conforms to the Gaussian distribution. Thus, for a pixel p belonging to a moving object, the probability distribution of its intensity value I_p can be formulated as

$$P(I_p) = \eta(I_p, \mu_1, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu_1)^2}{2\sigma^2}\right) \quad (10)$$

where η is a Gaussian probability function and μ_1 and σ are the mean and standard deviation of the Gaussian function, respectively. Subsequently, we employ ℓ_2 regularization to encode the temporal continuity prior constraint for moving objects.

C. MOD Based on LRSD Framework

1) Problem Definition: Based on the above analyses, we integrate E-3DTV regularization and Gaussian prior into the LRSD framework for MOD in satellite videos, which is formulated as follows:

$$\begin{aligned} &\min_{\mathbf{B},\mathbf{T}} \|\mathbf{B}\|_{\text{E-3DTV}} + \lambda \|\mathbf{T}\|_2 \\ &\text{s.t. } \mathbf{O} = \mathbf{B} + \mathbf{T} \end{aligned} \quad (11)$$

where λ represents the weight balancing the effect of background and moving objects. Based on (9) and (11), the model can be formulated as

$$\begin{aligned} &\min_{\mathbf{B},\mathbf{U}_i,\mathbf{V}_i,\mathbf{T}} \sum_{i=1}^{3} \|\mathbf{G}_i\|_1 + \lambda \|\mathbf{T}\|_2 \\ &\qquad = \sum_{i=1}^{3} \tau \|\mathbf{U}_i\|_1 + \lambda \|\mathbf{T}\|_2 \\ &\text{s.t. } \mathbf{O} = \mathbf{B} + \mathbf{T} \\ &\quad \nabla_i \mathbf{O} = \mathbf{U}_i \mathbf{V}_i^{\mathrm{T}}, \mathbf{V}_i^{\mathrm{T}} \mathbf{V}_i = \mathbf{I} \\ &\quad \mathbf{U}_i \in \mathbb{R}^{HW \times r}, \quad \mathbf{V}_i \in \mathbb{R}^{L \times r} \end{aligned} \quad (12)$$

where ∇ is the gradient operator and τ is the the sparsity basis coefficient. Since the size of $\mathbf{U}$ is smaller than $\mathbf{B}$, our method obtains a faster detection speed than most existing LRSD-based methods.

2) Optimization: According to (12), this minimization problem can be considered as the sum of two functions with uncoupled variables under a two-block linear constraint and can be optimized through ADMM [38], [39]. The main idea of ADMM optimization method is to minimize the dual form

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of original constrained optimization problem over one variable while fixing other variables at each iteration and then repeat this process alternatively with increasing positive penalty scalar μ until it converges. In our problem, the augmented Lagrangian function of (12) is given by

$$\begin{aligned}\mathcal{L}_\mu(\mathbf{B},\mathbf{T},\mathbf{U}_i,\mathbf{V}_i,\mathbf{G}_i,\mathbf{M}_i,Y)&=\sum_{i=1}^{3}\tau\|\mathbf{U}_i\|_1+\lambda\|\mathbf{T}\|_2\\&+\sum_{i=1}^{3}\langle\mathbf{M}_i,\mathbf{G}_i-\mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}\rangle+\frac{\mu}{2}\left\|\mathbf{G}_i-\mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}\right\|_F^2\\&+\langle\mathbf{Y},\mathbf{O}-\mathbf{B}-\mathbf{T}\rangle+\frac{\mu}{2}\|\mathbf{O}-\mathbf{B}-\mathbf{T}\|_F^2\end{aligned}\tag{13}$$

where $\mathbf{Y}$ and $\mathbf{M}_i$ are Lagrange multipliers, and μ is a positive penalty scalar in the ADMM algorithm.

In the ADMM framework, we need to alternatively optimize each variable involved in (13) while keeping all others fixed. We then discuss how to solve its subproblems with respect to each involved variable.

a) Update $\mathbf{B}$*:* Deriving (13) with respect to $\mathbf{B}$, we can obtain the following linear system:

$$\begin{aligned}&\left(\mu\mathbf{I}+\mu\sum_{i=1}^{3}\nabla_i^{\mathrm{T}}\nabla_i\right)(\mathbf{B})\\&\quad=\mu(\mathbf{O}-\mathbf{T})+\mathrm{Y}+\mu\sum_{i=1}^{3}\nabla_i^{\mathrm{T}}\left(\mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}\right)-\nabla_i^{\mathrm{T}}(\mathbf{M}_i)\end{aligned}\tag{14}$$

where $\nabla_i^{\mathrm{T}}(\cdot)$ indicates the transposition operator of $\nabla_i(\cdot)$. Specifically, by performing Fourier transform on both sides of (14) and adopting the convolution theorem, the closed-form solution to $\mathbf{B}$ is obtained according to [40]

$$\begin{cases}\mathbf{H}_b=\displaystyle\sum_{i=1}^{3}\mathcal{F}(\mathbf{D}_i)^*\odot\mathcal{F}\left(\mathrm{fold}\left(\mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}\right)-\mathbf{M}_i\right)\\ \mathbf{Q}_b=|\mathcal{F}(\mathbf{D}_1)|^2+\mathcal{F}(\mathbf{D}_2)|^2+|\mathcal{F}(\mathbf{D}_3)|^2\\ \mathbf{B}=\mathcal{F}^{-1}\left(\dfrac{\mathcal{F}(\ \mathrm{fold}\ (\mu(\mathbf{O}-\mathbf{T})+\mathbf{Y}))+\mathbf{H}_b}{\mu\mathbf{1}+\mu\mathbf{Q}_b}\right)\end{cases}\tag{15}$$

where $\mathbf{1}$ represents the tensor with all elements being 1; $\odot$ is the element-wise multiplication; $\mathbf{D}_i$ is the correlated difference filter; $\mathcal{F}^1(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ are Fourier transform and its inverse transform, respectively; and $|\cdot|^2$ is the element-wise square operation.

b) Update $\mathbf{U}_i$*:* Extracting all items containing $\mathbf{U}_i$ from (13), we obtain

$$\mathbf{U}_i=\underset{\mathbf{U}_i}{\mathrm{argmin}}\tau\|\mathbf{U}_i\|_1+\frac{\mu}{2}\left\|\mathbf{U}_i-\left(\mathbf{G}_i+\frac{\mathbf{M}_i}{\mu}\right)\mathbf{V}_i\right\|_F^2.\tag{16}$$

The subproblem (16) can be calculated in the closed form as follows:

$$\mathbf{U}_i=\mathcal{S}_{\frac{\tau}{\mu}}\left(\left(\mathbf{G}_i+\frac{\mathbf{M}_i}{\mu}\right)\mathbf{V}_i\right)\tag{17}$$

where $\mathcal{S}_\Delta$ is the soft-thresholding operator [41] and Δ is the threshold value, which must be greater than 0. The definition of $\mathcal{S}_\Delta$ is as follows:

$$\mathcal{S}_\Delta(\mathbf{x})=\begin{cases}x-\Delta, & \text{if } \mathbf{x}>\Delta\\ x+\Delta, & \text{if } \mathbf{x}<-\Delta\\ 0, & \text{otherwise .}\end{cases}\tag{18}$$

c) Update $\mathbf{V}_i$*:* Extracting all items containing orthogonal $\mathbf{V}_i$ from (13), we have

$$\begin{aligned}\mathbf{V}_i&=\underset{\mathbf{V}_i}{\mathrm{argmin}}\frac{\mu}{2}\left\|\mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}-\left(\mathbf{G}_i+\frac{\mathbf{M}_i}{\mu}\right)\right\|_F^2\\&=\underset{\mathbf{V}_i}{\mathrm{argmin}}\left\langle\left(\mathbf{G}_i+\frac{\mathbf{M}_i}{\mu}\right)^{\mathrm{T}}\mathbf{U}_i,\mathbf{V}_i\right\rangle.\end{aligned}\tag{19}$$

According to theorem [42], the global solution of subproblem (19) can be calculated in the closed form as follows:

$$\begin{cases}[\mathbf{A},\mathbf{C},\mathbf{E}]=\mathrm{svd}\left(\left(\mathbf{G}_i+\dfrac{\mathbf{M}_i}{\mu}\right)^{\mathrm{T}}\mathbf{U}_i^k\right)\\ \mathbf{V}_i^{=}\mathbf{A}\mathbf{E}^{\mathrm{T}}.\end{cases}\tag{20}$$

d) Update $\mathbf{T}$*:* Extracting all items containing orthogonal $\mathbf{T}$ from (13), we can get

$$\begin{aligned}\mathbf{T}&=\underset{\mathbf{T}}{\mathrm{argmin}}\lambda\|\mathbf{T}\|_2+\frac{\mu}{2}\left\|\mathbf{O}-\mathbf{B}-\mathbf{T}+\frac{1}{\mu}\mathbf{Y}\right\|_F^2\\&=\frac{\mu}{2\lambda+\mu}\left(\mathbf{O}-\mathbf{B}+\frac{1}{\mu}\mathbf{Y}\right).\end{aligned}\tag{21}$$

Based on the general ADMM principle, the multipliers are further updated by the following equations:

$$\begin{cases}\mathbf{M}_i=\mathbf{M}_i+\mu\left(\mathbf{G}_i-\mathbf{U}_i\mathbf{V}_i^{\mathrm{T}}\right)\\ \mathbf{Y}=\mathbf{Y}+\mu(\mathbf{O}-\mathbf{B}-\mathbf{T}).\end{cases}\tag{22}$$

The entire algorithm of optimizing (13) is summarized in Algorithm 1.

Algorithm 1 Optimization of MOD in Satellite Videos

Require: The satellite video $\mathcal{O}\in\mathbb{R}^{H\times W\times L}$, folding to matrix $\mathbf{O}\in\mathbb{R}^{HW\times L}$, $\lambda>0$, $\mu>0$, TV unfolding matrix rank:γ, $\overline{\mu}=\mu\times10^{12}$, $\rho=1.5$, stopping criteria $\epsilon=$ 1e-3 and *maxIter* = 400.

Ensure: $\mathbf{B}$ and $\mathbf{T}$.

$\mathbf{B}^0=\mathbf{T}^0=\mathbf{U}_i^0=\mathbf{V}_i^0=\mathbf{M}_i^0=\mathbf{Y}^0=\mathbf{0}\in\mathbb{R}^{HW\times L}$.
$k=0$
while not converged **do**
  Update $\mathbf{B}$ by (15)
  Update $\mathbf{U}_i$ by (17)
  Update $\mathbf{V}_i$ by (20)
  Update $\mathbf{T}$ by (21)
  Update the $\mathbf{M}_i$ and $\mathbf{Y}$ by (22)
  Update μ by $\mu=\min(\rho\mu,\overline{\mu})$
  Check the convergence conditions $\frac{\|\mathbf{O}-\mathbf{B}-\mathbf{T}\|_F^2}{\|\mathbf{O}\|_F^2}\le\epsilon$
end while
return $\mathbf{B}\in\mathbb{R}^{HW\times L}$ and Unfold $\mathbf{T}\in\mathbb{R}^{HW\times L}$

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TABLE I
DETAILS OF THE EVALUATION DATASETS

Dataset	Name	Frame size × # Frames
VISO	Video 001 ∼ Video 006	1024×1024×325
	Video 007	1024×1024×300
SkySat	Video 008	400×400×700
	Video 009	600×600×700

V. EXPERIMENTAL RESULTS

In this section, we conduct extensive experiments to evaluate the MOD performance of the proposed method on both SkySat dataset [9] and VISO dataset [16].

A. Experimental Setups

1) Datasets: In our experiments, nine satellite videos from the VISO dataset and SkySat dataset are used, and the details are shown in Table I.

a) VISO Dataset: This dataset comes from the Jilin-1 satellite constellation developed by China Changchun Satellite Technology Company Ltd.[1] The VISO dataset is free for scientific research and can be downloaded from the website[2]. Seven videos (Videos 001∼007) with colors from the test dataset are used in our experiments, and the spatial resolution of these videos is 1 m at 10 FPS. Two satellite videos (Videos 002 and 003) contain relatively less complex backgrounds, and the others contain challenging backgrounds with content changes and uneven moving. In these videos, most of the moving objects are motor vehicles with small sizes and partial occlusions.

b) SkySat Dataset: This dataset is captured by the SkySat-1 satellite constellation shared by SkyBox[3] imaging over Las Vagas, USA, on March 25, 2014. Both the satellite videos and the ground truth are publicly available on the website.[4] The spatial resolution of these videos is approximately 1 m, and the frame rate is 30. Two grayscale satellite videos with 700 frames (Videos 008 and 009) are used in our experiments. In addition, Video 008 contains complex and moving backgrounds, while Video 009 is mainly composed of roads and less complex backgrounds. Video 008 contains totally 27 473 vehicles with 39 objects per frame on average and a speed of 1.38 pixels per frame, while Video 009 contains 52 805 vehicles with 75 objects per frame on average and a speed of 1.19 pixels per frame.

2) Implementation Details: According to [37], we set the sparsity basis coefficient $\tau = c\sqrt{\mathrm{HW}}$ in (13), where H and W are the spatial height and width of the satellite videos, respectively, and c is set to 0.0001 to guarantee a relatively good performance of our method in all datasets. Inspired by the parameter settings of RPCA [22], we set coefficient $\lambda = 1$. The value of rank r mainly describes the low-rank property of input satellite video and is related to computational complexity. Considering that the gradient maps along the spatial dimension and temporal dimension have different insightful structures, we impose different sparsity consideration across spatial (i.e., height and width) and temporal dimensions. Therefore, we set r_1 and r_2 to be 25 and 3 for the spatial and temporal dimensions, respectively. It should be noted that our proposed method does not encompass the specific procedures for data preprocessing and postprocessing. All experiments are performed on a computer with 128-GB RAM and 13th Gen Intel[5] Core[6] i9-13900KF @ 3.00 GHz. The operating system is Windows 10 with 64 bits, and the executive software is MATLAB 2021a.

B. Evaluation Metrics

In order to comprehensively evaluate the performance of MOD in terms of accuracy and efficiency, we follow [9], [10], [16] to use the precision, recall, and F1 score as the MOD evaluation metrics and also use the average time cost on each input image for efficiency.

The recall and precision are defined as follows:

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{23}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{24}$$

where FP, true positive (TP), and FN indicate the number of false alarms, the number of correct detections, and the number of missing detections, respectively. F_1 is computed as

$$F_1 = \frac{2 \times \text{recall} \times \text{precision}}{\text{recall} + \text{precision}}. \tag{25}$$

The F1 score balances the recall and precision and gives an overall quantitative evaluation. Due to the small size of moving objects in satellite videos, even minor shifts in the predicted bounding box can lead to significant fluctuations in the IoU value. Therefore, in this article, we follow [21] to consider a predicted bounding box as a TP if the distance between its center and the ground-truth center is less than a predefined threshold of 5 pixels.

C. Performance

To demonstrate the superiority of the proposed method, we compare it with 12 state-of-the-art detectors, including frame differencing-based methods (i.e., D&T [15] and MMB [16]), background modeling-based methods (i.e., DECOLOR [44], Godec [43], ClusterNet [17], DTTP [32], ELSD [9], B-MCMD [10], WSNM [29], and O-LSD [28]), and learning-based methods (i.e., DSFNet [18], ClusterNet [17], and UMOD [21]).

1) VISO Dataset: The quantitative evaluations conducted on the VISO dataset are shown in Table II. We can observe that the proposed method achieves the highest average precision performance among all background modeling-based methods. Compared with frame differencing-based method D&T, our method promotes the average F1 score from 79% to 83%. Considering the characteristics of complex dynamic backgrounds

[1]http://www.jl1.cn/index.aspx
[2]https://github.com/TheLearningAndVisionAtelierLAVA/VISO
[3]https://earth.esa.int/eogateway/missions/skysat
[4]https://github.com/zhangjunpeng9354/satellite video mod groundtruth.git
[5]Trademarked.
[6]Registered trademark.

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TABLE II

RECALL (RE), PRECISION (PR), AND $F1$ RESULTS OBTAINED BY DIFFERENT METHODS ON SEVEN SATELLITE VIDEOS OF THE VISO DATASET. TIME COST (S) FOR A SINGLE IMAGE OF DIFFERENT METHODS IS ALSO LISTED IN THE TABLE. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED, RESPECTIVELY

Method	Video 001			Video 002			Video 003			Video 004			Video 005			Video 006			Video 007			Average			Time Cost
	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	Re↑	Pr↑	F1↑	(s)
DSFNet [18]	**91%**	92%	**92%**	**88%**	85%	**86%**	**95%**	81%	88%	**82%**	91%	**86%**	**95%**	67%	79%	**81%**	86%	**83%**	83%	**87%**	**85%**	**88%**	84%	**86%**	0.29
ClusterNet [17]	74%	63%	68%	61%	75%	67%	81%	65%	72%	47%	68%	56%	74%	79%	76%	73%	68%	71%	83%	64%	72%	71%	69%	69%	0.40
UMOD [21]	83%	90%	86%	76%	88%	81%	90%	88%	89%	65%	83%	73%	72%	89%	80%	73%	86%	79%	83%	74%	82%	77%	85%	81%	0.48
D&T [15]	70%	90%	79%	67%	82%	74%	82%	81%	82%	72%	79%	76%	61%	77%	68%	62%	72%	67%	84%	39%	53%	71%	75%	71%	0.18
MMB [16]	80%	94%	86%	71%	88%	79%	85%	90%	88%	74%	81%	78%	68%	83%	75%	66%	84%	74%	**86%**	66%	74%	76%	84%	79%	0.50
Godec [43]	84%	77%	80%	78%	78%	78%	72%	77%	75%	63%	78%	70%	77%	71%	74%	67%	65%	66%	27%	38%	32%	67%	69%	68%	5.10
DECOLOR [44]	36%	**99%**	53%	80%	81%	81%	90%	78%	83%	43%	**99%**	60%	79%	75%	77%	78%	64%	71%	33%	70%	45%	63%	81%	67%	8.20
DTTP [32]	74%	48%	58%	66%	80%	72%	73%	82%	77%	55%	58%	56%	53%	68%	60%	53%	68%	60%	23%	47%	30%	57%	64%	59%	6.36
E-LSD [9]	71%	76%	73%	64%	39%	48%	75%	87%	81%	57%	87%	69%	61%	82%	70%	53%	75%	62%	59%	58%	58%	63%	72%	66%	94.20
B-MCMD [10]	75%	91%	82%	71%	80%	75%	79%	74%	76%	67%	68%	68%	53%	76%	63%	63%	65%	64%	77%	41%	53%	69%	71%	69%	252.70
WSNM [29]	84%	70%	77%	79%	85%	82%	91%	87%	89%	65%	87%	74%	76%	85%	80%	72%	83%	77%	73%	77%	75%	77%	82%	79%	3.12
Ours	78%	97%	86%	79%	**90%**	84%	90%	**91%**	**91%**	67%	93%	78%	75%	**90%**	**82%**	74%	**90%**	81%	77%	75%	76%	77%	**89%**	83%	0.95

and slow-moving objects, we use the E-3DTV regularization term and Gaussian prior to model background and moving objects, respectively, which makes our method improve detection performance with reduced false alarms. In contrast, other background modeling-based methods suffer significant performance degradation when applied to slow-moving objects in complex dynamic background scenes in satellite videos. In addition, because of the high similarity of motion characteristics between the short-distance moving objects and the stationary background objects with local misalignment, our method performs poor compared with the learning-based method DSFNet. However, the better performance of DSFNet depends on the high-quality annotated large-scale dataset and also needs two weeks to train the model. In contrast, our method obtains competitive performance with much less training time and does not require a high-quality annotated dataset.

Fig. 3 shows the visualization results of different methods on two frames of VISO dataset. The first frame is the 128th frame in Video 001, which mainly contains the changing background of the sea. The second frame is the 45th frame in Video 005, which mainly contains the moving vehicles on wild road. From the results shown in the first two rows, we can observe that our method is robust to complex dynamic background compared with existing methods. Due to the complex changing backgrounds, current methods generate a large number of false alarms. Because of the effective explorations of the spatial–temporal information in video sequences, our method improves the robustness to the background variances. From the results shown in the last two rows, we can observe that our method obtains reliable detection results with fewer false alarms (red bounding boxes), while existing methods exhibit a high ratio of missing detections (blue bounding boxes) in the scenario with densely packed objects.

The efficiency comparison between the proposed method and other methods is listed in the last column of Table II, which is computed as the average time cost on the input image with a size of 1024×1024. Compared with other background modeling-based methods, our method achieves at least $3\times$ acceleration. This better efficiency is that our method imposes regularization constraint on the gradient maps rather than the original video sequences.

2) SkySat Dataset: We also evaluate the performance of the proposed method on the SkySat dataset. For learning-based methods, our method first trains the network on the VISO dataset and then tests on the SkySat dataset. As shown in Table III, our method achieves the highest average precision with the score of 90% and average F_1 with the score of 91% compared with all other the methods. Meanwhile, the average recall of our method is only relatively lower than the best performance. Compared with frame differencing-based methods, our method obviously outperforms it in terms of all metrics. Benefiting from the focus on exploring the characteristics of background and moving objects in satellite videos, our method can improve detection performance with reduced false alarms. In contrast, other background modeling-based methods suffer significant performance degradation when applied to moving objects in complex dynamic background scenes. In addition, due to the significant differences in data between the two datasets and lack of sufficient training data, our method obtains competitive performance compared with learning-based methods and is more adaptable to various scenarios. The efficiency comparison between our method and other methods is listed in the last column of Table III, which is computed as the average time cost on the input image. From Table III, we can observe that our method also achieves the best efficiency compared with all other methods.

Fig. 4 visualizes the detection results on two frames of the SkySat dataset. The first frame is the 210th frame in Video 008, which mainly contains the local misalignment caused by the satellite platform motion. The second frame is the 345th frame in Video 009, which mainly contains

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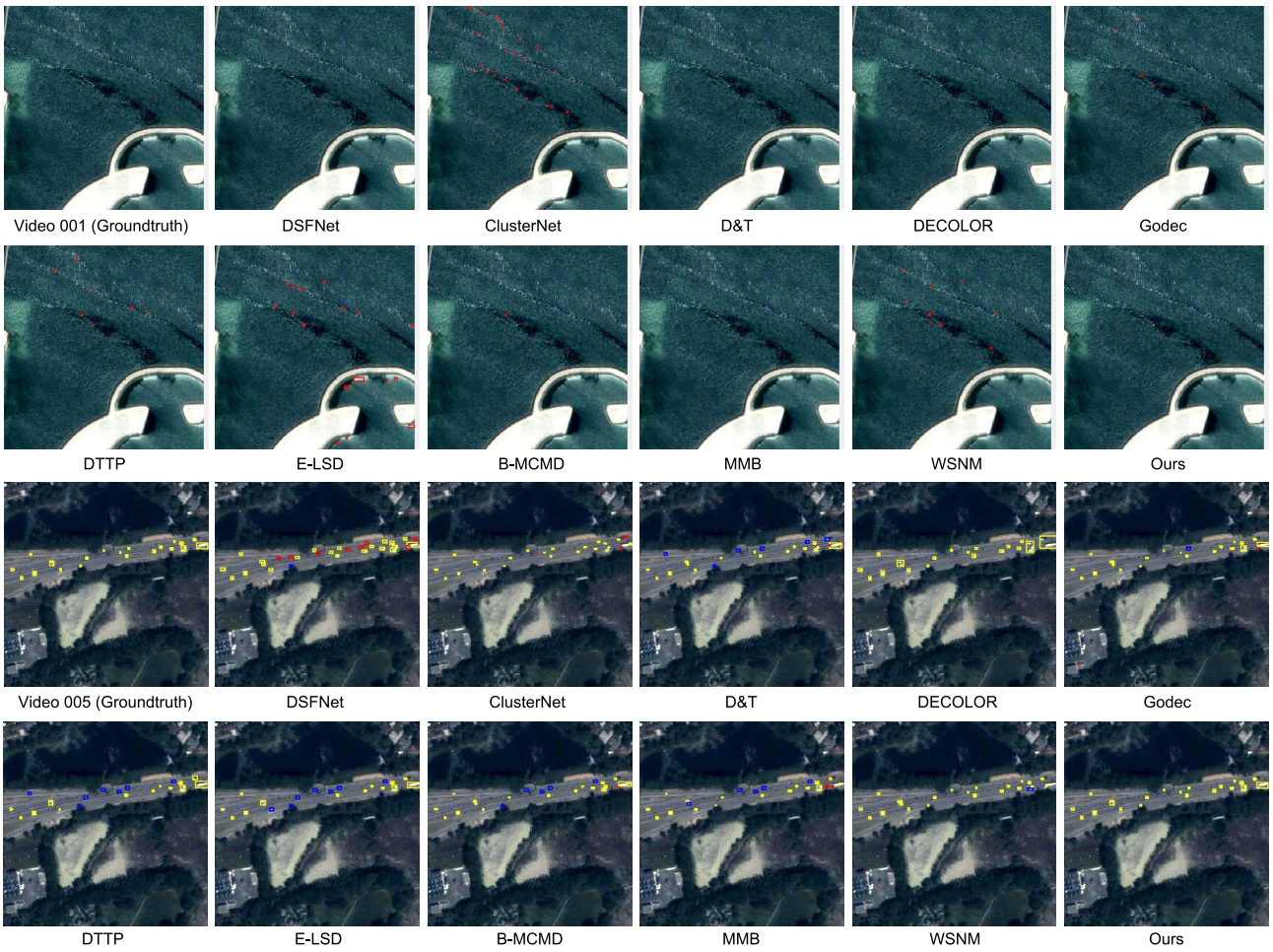


Fig. 3. Comparison of visualization results of different methods on changing background and densely packed objects. The yellow rectangle represents the correctly detected objects, the red rectangle represents the false alarms, and the blue rectangle represents the miss detections.

TABLE III

RECALL, PRECISION, AND $F1$ RESULTS OBTAINED BY DIFFERENT METHODS ON TWO SATELLITE VIDEOS OF THE SKYSAT DATASET. TIME COST (S) FOR A SINGLE IMAGE OF DIFFERENT METHODS IS ALSO LISTED IN THE TABLE. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN BOLD AND UNDERLINED

Method	Video 008			Video 009			Average			Time Cost (s)
	Recall ↑	Precision ↑	F1 ↑	Recall↑	Precision↑	F1 ↑	Recall↑	Precision↑	F1 ↑	
DSFNet [18]	88%	65%	75%	90%	78%	84%	89%	71%	79%	0.29
ClusterNet [17]	77%	46%	57%	79%	58%	66%	78%	52%	62%	0.40
D&T [15]	78%	59%	67%	75%	85%	80%	76%	72%	73%	0.12
MMB [16]	68%	82%	75%	63%	84%	72%	66%	83%	73%	0.38
Godec [42]	**95%**	36%	52%	90%	81%	85%	93%	59%	69%	0.49
DECOLOR [44]	77%	59%	67%	79%	81%	80%	78%	70%	73%	0.72
DTTP [32]	80%	67%	73%	74%	94%	83%	77%	80%	78%	0.60
E-LSD [9]	85%	79%	82%	80%	94%	86%	82%	86%	84%	31.20
O-LSD [34]	65%	64%	64%	73%	90%	87%	78%	76%	72%	8.70
B-MCD [10]	82%	84%	85%	80%	**95%**	87%	81%	90%	86%	92.41
WSNM [29]	94%	76%	84%	**95%**	80%	87%	**94%**	78%	86%	0.33
Ours	92%	**88%**	**90%**	92%	92%	**92%**	92%	**90%**	**91%**	0.19

the moving vehicles on wild road. The results in the first two rows indicate that our method obtains greater robustness against complex dynamic backgrounds compared with existing methods. Current methods tend to produce a high number

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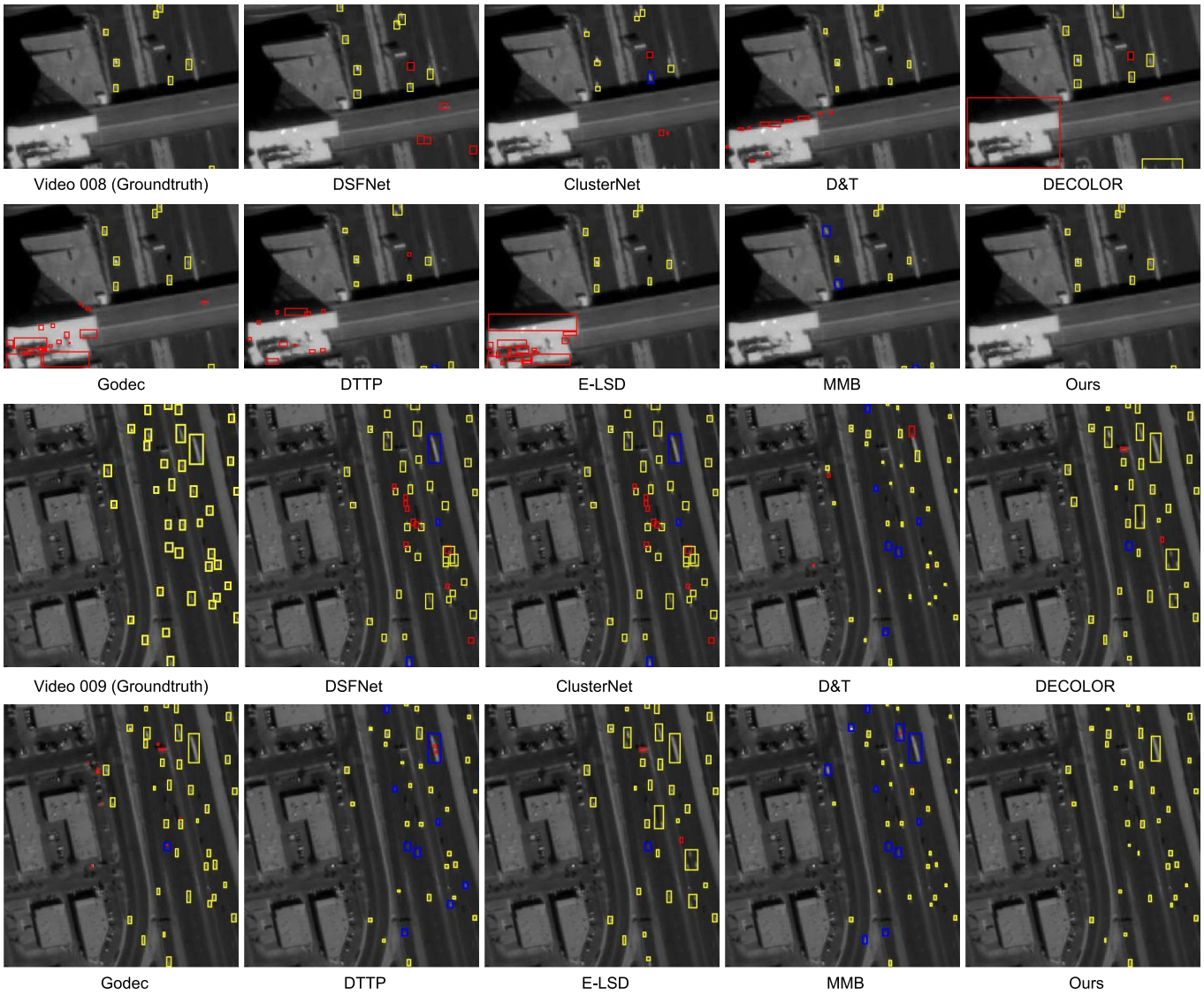


Fig. 4. Comparison of visualization results of different methods on a frame of the SkySat dataset. The yellow rectangle represents the correctly detected objects, the red rectangle represents the false alarms, and the blue rectangle represents the miss detection.

TABLE IV

RECALL, PRECISION, AND $F1$ RESULTS OBTAINED BY DIFFERENT METHODS ON THE VISO DATASET AND THE SKYSAT DATASET. THE BEST RESULTS ARE SHOWN IN BOLDFACE

Model	Configuration		Dataset							
	Low-rank Background Modeling	Moving Object Modeling	VISO				SkySat			
			Recall	Precision	F1	FPS	Recall	Precision	F1	FPS
A	3DTV	Gaussian Prior	78%	60%	67%	0.05	68%	91%	78%	0.22
B	E-3DTV	Laplacian Prior	**82%**	79%	80%	0.53	78%	**94%**	85%	2.63
Ours	E-3DTV	Gaussian Prior	77%	**89%**	**83%**	**1.1**	**92%**	90%	**91%**	**4.98**

of false alarms due to the variability of these changing backgrounds. In contrast, our method effectively harnesses the spatial–temporal information from video sequences, enhancing its resilience to background variations. Furthermore, the results in the last two rows reveal that our method achieves reliable detections with fewer false alarms (red bounding boxes). On the other hand, existing methods show a significantly higher rate of missed detections (blue bounding boxes) in scenarios containing densely packed objects.

D. Ablation Studies

In this section, we analyze the effectiveness of the proposed modules on the VISO and SkySat datasets, including

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TABLE V

EVALUATION ON THE SYNTHETIC DATA. PSNR, RECALL, PRECISION, AND $F1$ ARE USED FOR PERFORMANCE COMPARISON. THE BEST RESULTS ARE SHOWN IN BOLDFACE

Methods	PSNR	Recall	Precision	F1
DECOLOR [44]	26.42	48.9%	**99.4%**	65.7%
E-LSD [9]	26.37	68.9%	79.5%	73.8%
B-MCMD [10]	26.52	77.0%	79.0%	78.0%
WSNM [29]	30.20	79.5%	99.1%	88.2%
UMOD [21]	29.42	**94.2%**	79.2%	86.0%
Ours	**31.55**	88.0%	95.3%	**91.5%**

TABLE VI

RECALL, PRECISION, AND $F1$ RESULTS OBTAINED BY DIFFERENT COMBINATIONS OF RANK ON VIDEO 008. THE BEST RESULTS ARE SHOWN IN BOLDFACE

r_1	r_2	Recall	Precision	F1	r_1	r_2	Recall	Precision	F1
25	3	**92.4%**	85.9%	**89.0%**	5	3	**94.6%**	53.7%	68.5%
25	5	81.4%	90.1%	85.9%	15	3	93.4%	79.1%	85.7%
25	7	78.9%	91.6%	84.6%	25	3	92.4%	85.9%	**89.0%**
25	9	75.6%	92.0%	83.0%	35	3	81.8%	90.6%	86.0%
25	11	72.1%	92.6%	81.1%	45	3	76.6%	**91.6%**	83.4%
25	25	62.2%	**93.7%**	74.8%	55	3	75.0%	81.4%	82.3%

background modeling based on E-3DTV and moving object modeling based on Gaussian prior. The results of the ablation experiments are shown in Table IV. In addition, we also evaluate the background reconstruction capability of our method through simulation, and the results are shown in Table V. Finally, we evaluate the effect of different combinations of ranks r_1 and r_2, and the results are shown in Table VI.

1) Effectiveness of Background Modeling Based on E-3DTV: To validate the effectiveness of our background modeling method based on E-3DTV, we replace the background reconstruction model with the original 3DTV. Comparing model A with our model in Table IV, we can observe that our method achieves better $F1$ scores on the VISO dataset and the SkySat dataset and promotes the $F1$ metric about 16% and 13% for these two datasets, respectively. In our method, we assume that the gradient maps on the temporal and spatial dimensions conform to different physical meanings and then model these two kinds of gradient maps with different low-rank characteristics in 3DTV. However, the original 3DTV just treats the gradient maps on the temporal and spatial dimensions the same and thus cannot capture the inherent underlying structure of the satellite videos. Additionally, comparing the FPS metric in model A with our model, we can also observe that our method also achieves the best computational efficiency on both the VISO and SkySat datasets. This is because model A directly applies the original 3DTV on the gradient maps without LRMF, while our method applies the 3DTV on the factorized low-rank matrix.

To further evaluate the background reconstruction capability of our method, we simulate moving objects on the clean background and then apply different methods for background reconstruction. In addition to the MOD metrics, following [45], we also use PSNR calculated between the reconstructed background and the ground-truth one as the quantitative metric for reconstruction performance evaluation. Specifically, we compare our method with five background modeling-based methods, including DECOLOR [44], E-LSD [9], B-MCMD [10], WSNM [29], and UMOD [21]. The quantitative comparison results are shown in Table V. We can observe that our method achieves the best PSNR and F1 scores, which demonstrates the effectiveness of our background modeling method.

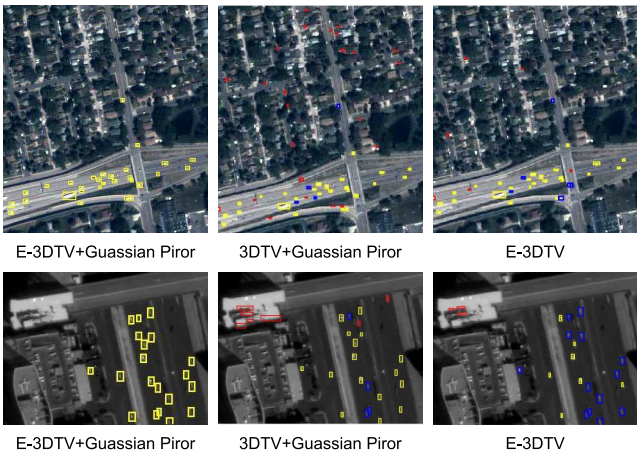


Fig. 5. Visualization of detection results for ablation studies on two frames from the VISO and SkySat datasets. The yellow rectangle represents the correctly detected objects, the red rectangle represents the false alarms, and the blue rectangle represents the miss detections.

2) Effectiveness of Moving Object Modeling Based on Gaussian Prior: To validate the effectiveness of moving object modeling based on Gaussian distribution prior, we utilize the Gaussian distribution prior to replace Laplacian distribution prior to encode the temporal continuity prior constraints for moving objects. Comparing model B with our method in Table IV, we can observe that our method promotes the F1 metric about 3% and 6% for the VISO datasets and SkySat datasets, respectively. The reason for better performance of our method is that the intensity changing characteristic of the moving objects across frames exhibits a Gaussian prior, while other methods just utilize the sparsity characteristics of moving objects in each frame alone. Therefore, benefiting from the ℓ_2 regularization, our method exhibits stronger detection ability than ℓ_1 regularization in complex scenes.

Fig. 5 visualizes the detection results on two frames from the VISO and SkySat datasets. The first frame is the 48th frame in Video 005, which mainly contains the moving vehicles on the wild road. The second frame is the fifth frame in Video 008, which contains the local misalignment caused by the satellite platform motion. The results in the first row show that our method is more robust against complex backgrounds compared to existing methods. While current methods often trigger many false alarms, our method still effectively obtains superior detection performance due to the advantages of proposed background modeling and moving

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object modeling. Additionally, the results in the second row demonstrate that our method provides reliable detections with fewer false alarms when dealing with local misalignment. This is because the E-3DTV regularization models the temporal and spatial gradient maps with different low-rank characteristics, while the original 3DTV just treats the gradient maps on the temporal and spatial dimensions the same. Thus, the effect of background moving caused by the local misalignment cannot be effectively eliminated using the original 3DTV.

3) Effectiveness of Different Rank Combinations: In this section, we conduct experiments on Video 008 from the SkySat dataset to evaluate the effectiveness of different combinations of ranks r_1 and r_2. The parameter r_1 reflects the ability to separate moving objects from the background within individual frames. Specifically, r_1 is related to the rank of the background model for each frame. The rank of a matrix indicates the number of linearly independent components or features that contribute to the representation of the background. A higher r_1 means more complex background modeling, potentially including moving objects as part of the background. Conversely, a lower r_1 value may result in a simpler background model that does not capture all the variations in the scene, potentially causing underfitting and misclassification of moving objects as background. The parameter r_2 controls the rank of the background model across multiple frames in a satellite video. It impacts the complexity of background variations over time and indicates the model's capacity to capture these variations. A higher r_2 value means that the model is more sensitive to changes, potentially including noise and moving objects, which can decrease reconstruction accuracy. As shown in Table VI, r_1 ranges from 5 to 55 and r_2 ranges from 3 to 25. When fixing r_1 and increasing r_2 from 3 to 25, the performance of $F1$ decreases. This is because more noises and moving objects are treated as the variations of the background across frames in the satellite videos as r_2 increases, making the background reconstruction not accurate enough. When fixing r_2 and increasing r_1 from 5 to 55, the performance of $F1$ also decreases. This is because moving objects are treated as the background in each frame in satellite videos as r_1 increases, also making the background reconstruction not accurate enough. Through experiments, we observe that the best performance is achieved when $r_1 = 25$ and $r_2 = 3$.

E. Complexity Analyses

In this section, we analyze the computational complexity of our method. In each iteration, the most time-consuming steps are the updating of $\mathbf{B}$ and $\mathbf{V}_i$. Assuming that the size of the satellite video tensor is $\mathbf{B} \in \mathbb{R}^{HW \times L}$, the updating of $\mathbf{B}$ requires the FFT calculation, and its computational complexity is $O(HWL \log(HWL))$. In addition, the updating of $\mathbf{V}_i$ needs to implement three SVD calculations, and its computational complexity is $O(HWL \times (HW+HL+WL)+H^3+W^3+L^3)$. Thus, the total computational complexity of our method in each iteration is $O(HWL \times (HW + HL + WL) + H^3 + W^3 + L^3)$.

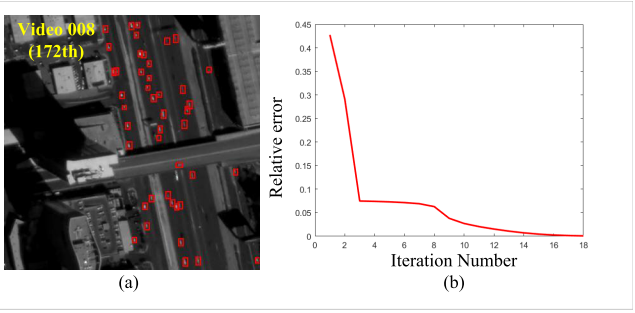


Fig. 6. (a) MOD results and (b) convergence curve obtained by our method on Video 008.

F. Convergence Analyses

Our method is a nonconvex method based on LRMF. Therefore, we use the convergence condition $\|\mathbf{O} - \mathbf{B} - \mathbf{T}\|_F^2 / \|\mathbf{O}\|_F^2 \leq \epsilon$ to analyze the convergence of our method. Specifically, the convergence experiments are performed on the 172th frame of Video 008 from the SkySat dataset. The MOD results are shown in Fig. 6(a), and the convergence curve is illustrated in Fig. 6(b). We can observe that as the number of iterations increases, the relative error swiftly approaches zero, demonstrating that our method achieves convergence quickly and attains optimal solutions efficiently.

VI. Conclusion

In this article, we propose an effective and efficient method for multiple object detection in satellite videos. We model the background with different Laplacian sparsity priors for the gradient maps along the spatial and temporal dimensions and model the intensity changing characteristics of moving objects across frames with Gaussian prior. By combining these two models into the LRSD framework, the proposed method achieves superior performance and high computational efficiency on the two public satellite video MOD datasets VISO and SkySat. However, our method is a traditional optimization method, and the heuristic knowledge about the data characteristics of satellite videos still limits its performance. In the future, we will explore integrating deep background priors with iterative optimization to develop an end-to-end network. In addition, we also consider learning the background with remote sensing fundamental models to further improve the detection performance.

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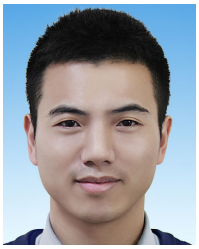
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